

Seat No: _____

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF IT & COMPUTER SCIENCE
MCA 2025-26 Mid-Term Examination

Semester: 3
 Subject Code: 05201231
 Subject Name: OSTF

Date: 07-08-2025
 Time: (1hr:30min)
 Total Marks: 40

Instructions:

1. Figures to the right indicate full marks.
2. Make suitable assumptions wherever necessary.

		CO	PO	BL	[10]
Q.1	Answer the following.				
(a)	10 MCQ questions of 1 mark each				
	1. What are the two main branches of Statistics? A) Qualitative and Quantitative Statistics B) Population and Sampling Statistics C) Hypothesis and Probability Statistics D) Descriptive and Inferential Statistics	2	2	Remembering (BT-1)	
	2. In the Pima Indians Diabetes Dataset, what is the dependent variable? A) Pregnancies B) Glucose concentration C) Outcome (0 or 1) D) BMI	2	2	Remembering (BT-1)	
	3. Which of the following measures are considered measures of centrality? A) Mean C) Mode B) Median D) All	2	2	Remembering (BT-1)	
	4. Which sampling method involves dividing the population into groups and selecting entire groups? A) Simple Random Sampling B) Stratified Sampling C) Cluster Sampling D) Systematic Sampling	2	2	Remembering (BT-1)	
	5. Which of the following are types of hypothesis tests? A) ANOVA C) Chi-square Test B) Z-test D) All	2	2	Remembering (BT-1)	
	6. The P-value indicates: A) Sample size B) Statistical significance C) Confidence interval D) Population parameter	2	2	Remembering (BT-1)	
	7. Which test is used to compare means from two different samples? A) One-sample t-test B) Two-sample t-test C) ANOVA D) Chi-square Test	2	2	Remembering (BT-1)	
	8. Which of the following are foundational rules of probability? A) Probability of Impossible Event is 0 B) Probability of Certain Event is 1 C) Addition Rule	1	2	Remembering (BT-1)	

	D) All				
	9. In statistics, degrees of freedom (DF) is calculated as: A) $N - 1$ C) $2N$ B) $N + 1$ D) $N/2$	1	2	Remembering (BT-1)	
	10. Conditional probability is denoted as: A) $P(A \text{ or } B)$ B) $P(A \text{ and } B)$ C) $P(A B)$ D) $P(B)$	1	2	Remembering (BT-1)	
					[12]
Q.2	Answer the following. (3 mark questions) Any four		1		
	Find the Error and correct code			Analyse (BT-4)	
	Q1. What is the error in the following code? numbers = [1, 2, 3, 4, 5] for i in range(0, len(numbers)): print(numbers[i]) print("Done") Error:	2			
	Q2. What is the issue in this function call? def add(a, b): return a + b result = add(5) print(result) Error:	2	1	Analyse (BT-4)	
	Q3. Find the error in the following list operation. my_list = [10, 20, 30] print(my_list[3]) Error:	2	1	Analyse (BT-4)	
	Q4. What is the error in this import statement? import numpy as np arr = np.array([1, 2, 3]) print(arr) Error:	2	1	Analyse (BT-4)	
	Q5. Find the mistake in this variable naming and assignment. 2value = 10 print(2value) Error:	2	1	Analyse (BT-4)	
Q.3	Attempt any TWO.				[8]
	(i) Explain how you would apply data preprocessing steps to prepare raw data for analysis. What are the common techniques you would use, and why?	1	2	Apply (BT-3)	(05)

	(ii) Explain perception and Actuation in Context of Data Science	1	2	Analyse (BT-4)	(05)
	(iii) Explain in Details How Hypothesis will Help In Data Science Engineering field.	1	2	Evaluate (BT-5)	(05)
Q.4	Answer the following.				[10]
(a)	Program: Calculate the factorial of a number provided by the user.	2	4	Apply (BT-3)	(05)
(b)	Program: Check if a given number is even or odd. OR	2	4	Analyze(BT-4)	(05)
(b)	Program: Create Grade sheet of students based on their marks.	2	4	Create (BT6)	(05)